

Physics Department
Fisika Departement

Semester Test 2

Semestertoets 2

FSK116

17 May 2014 / 17 Mei 2014

Surname and initials

Van en voorletters

Student number

Studentenommer

Signature

Handtekening

Time

2½ hours

Total

65 marks

Tyd

2½ ure

Totaal

65 punte

Internal Examiner:

Dr J M Nel

External Examiner

Mr R Q Odendaal

Interne Eksaminator

Eksterne Eksaminator

- Answer all questions in the space provided in PEN.
- Clearly show all calculations. Do not just write down the final answer.
- Clearly state all assumptions, rules and laws used.
- Where applicable draw the free-body diagrams.
- Demonstrate clearly that you understand the physics that you use to answer the question. This means that a numerically correct answer will not necessarily be awarded full marks.
- Programmable calculators may NOT be used.
- **NO** textbooks allowed.
- A cheat sheet may be used, but must adhere to the rules printed on it.
- All test and examination regulations of the University of Pretoria are applicable.

- *Beantwoord alle vrae in die spasie gelaat in PEN.*
- *Wys al jou berekeninge. Moenie net 'n antwoord neerskryf nie.*
- *Stel duidelik alle aannames, reëls en wette wat u gebruik.*
- *Waar van toepassing, teken die vryliggaamsdiagram.*
- *Wys dat jy die fisika wat jy gebruik om die vraag te beantwoord verstaan. Dit beteken dat volpunte sal nie noodwendig vir 'n numeries-korrekte antwoord gegee word nie.*
- *Programmeerbare sakrekenaars MAG NIE gebruik word nie.*
- ***GEEN** handboeke word toegelaat nie.*
- *'n Formuleblad mag gebruik word onderhewig aan die reëls daarop gedruk.*
- *Alle toets- en eksamenregulasies van die Universiteit van Pretoria is van toepassing.*

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Venue

Lokaal

Question 1/Vraag 1

A fly collides with the windscreen of a moving bus.

'n Vlieg tref die voorruit van 'n bewegende bus.

- a) Which of the two experiences a greater force due to the collision, the fly or the bus? Explain your answer.

Watter een van die twee ondervind die groter krag as gevolg van die botsing, die vlieg of die bus? Verduidelik jou antwoord.

[2]

- b) Which of the two experiences the greater acceleration? Explain your answer.

Watter een van die twee ondervind die groter versnelling? Verduidelik jou antwoord.

[3]

The figure shows a *conical pendulum*, in which the bob (the small object at the lower end of the cord) moves in a horizontal circle at constant speed. (The cord sweeps out a cone as the bob rotates.) The bob has a mass of 0.040 kg, the string has length $L = 0.90$ m and has negligible mass, and the bob follows a circular path of circumference 0.94 m.

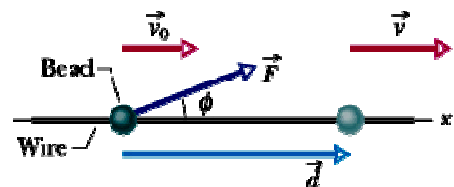
- [2+4]

- b) Determine the period of the motion.
Bepaal die periode van die beweging.

[5]

Question 3 / Vraag 3

Consider a bead, with a mass m that can slide along a frictionless wire that is stretched along a horizontal x axis. A constant force $\vec{F}$, directed at an angle ϕ with respect to the wire, accelerates the bead along the wire. Using Newton's second law and the associated acceleration, clearly show the relationship that exists between the work done by the force on the particle and the change in the kinetic energy of the particle. Explain each step and state any assumptions you may make.



Beskou 'n kraletjie, met massa m , wat op 'n wrywinglose draadjie, wat langs die horisontale x -as gespan is, gly. 'n Konstante krag, $\vec{F}$, wat teen 'n hoek ϕ met betrekking tot die draadjie gerig is, versnel die kraletjie langs die draadjie. Gebruik Newton se tweede wet, en die geassosieerde versnelling, om die verband tussen die arbeid verrig deur die krag op die deeltjie en die verandering in die kinetiese energie van die deeltjie te kry. Verduidelik elke stap en enige aannames wat jy maak.

[5]

A 2.5 kg block initially at rest, moves in a straight line on a horizontal frictionless surface under the influence of a force that varies with position of the block, as shown in the figure. Determine the speed of the block as it moves through the 8.0 m mark.

The graph shows the force exerted by a spring as a function of its position. The force is constant at 10 N for positions from 0 to 2 m. It then drops to 0 N at 2 m and remains at 0 N until 4 m. From 4 m to 6 m, the force is 0 N. Finally, from 6 m to 8 m, the force decreases linearly to -5 N.

5

Question 5 / Vraag 5

A 5.0 g marble is fired vertically upward using a spring gun. The spring must be compressed by 8.0 cm if the marble is to just reach a target 20 m above the marble's position **on the compressed** spring.

'n 5.0 g albaster word vertikaal opwaarts gevuur uit 'n veer-geweer. Die veer moet 8.0 cm saamgepers word sodat die albaster net-net die teiken bereik wat 20 m bo die albaster se posisie **op die saamgepersde** veer is.

- a) Determine the change in the gravitational potential energy of the marble-Earth system during the 20 m ascent.

Bepaal die verandering in die swaartekrag-potensiële energie van die albaster-Aarde stelsel tydens die 20 m opwaartse beweging.

[3]

- b) Determine the spring constant of the spring.

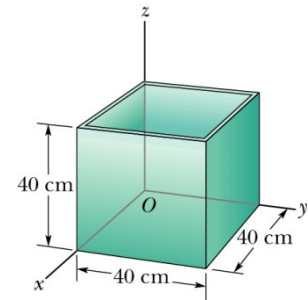
Bepaal die veerkonstante van die veer.

[5]

Question 6 / Vraag 6

A cubical box is constructed from uniform metal plate with negligible thickness. The box is open at the top and has an edge length $L = 40$ cm. Find the coordinates of the centre of mass of the box.

'n Kubieke boks word van uniforme metaal plaat met 'n weglaatbare dikte gemaak. Die boks is oop aan die bokant, en het 'n sylengte van $L = 40 \text{ cm}$. Bereken die koördinate van die massamiddelpunt van die boks.

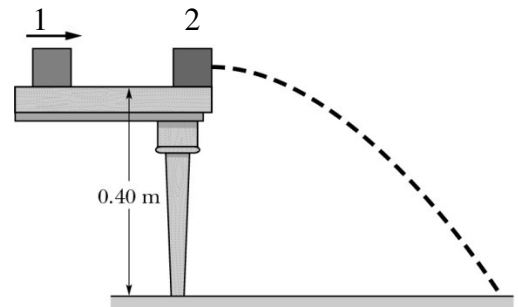


[9]

Question 7 / Vraag 7

A 3.2 kg box (1) slides on a horizontal frictionless table and collides with a second box (2) of 2.0 kg, initially at rest on the edge of the table, at a height $h = 0.40$ m. The two boxes collide and stick together due to sticky tape on their sides and together they hit the floor with a kinetic energy of 29 J. Determine the initial speed of the 3.2 kg box.

'n 3.2 kg boks (1) gly oor 'n horisontale wrywinglose tafel en bots met 'n tweede boks (2) van 2.0 kg, wat aanvanklik in rus is op die rant van die tafel, op 'n hoogte $h = 0.40$ m. Die twee bokse bots en klou vas aan mekaar as gevolg van kleefband op die kante van die bokse, en saam tref hulle die vloer met 'n kinetiese energie van 29 J. Bereken die aanvanklike spoed van die 3.2 kg boks.



[6]

Question 10 / Vraag 10

Trucks can be run on energy stored in a rotating flywheel, with an electric motor getting the flywheel up to its top speed of 100 rev/s. Such a flywheel can be a solid uniform cylinder with a mass of 500 kg and a radius of 1.0 m rotating about an axis along the length of the cylinder and through its centre of mass.

Motortrokke kan aangedryf word met energie wat gestoor word in 'n roterende vliegwiël, wat deur 'n elektriese motor gedraai word tot 'n maksimum spoed van 100 rev/s. So vliegwiël kan 'n soliede uniforme silinder met 'n massa van 500 kg en 'n radius van 1.0 m wees, wat roteer om 'n as langs die lengte van die silinder en deur die massa-middelpunt daarvan.

- a) Determine the kinetic energy of the flywheel after charging.

Bereken die kinetiese energie van die vliegwiël nadat dit gelaai is.

[3]

- b) If the truck uses an average power of 8.0 kW, for how many minutes can it operate before charging is necessary?

Indien die motortrok 'n gemiddelde drywing het van 8.0 kW, vir hoeveel minute kan die trok aangedryf word voor dit herlaai moet word?

[3]

Question 9 / Vraag 9

A uniform disk rotates at a constant angular velocity of 150 rev/min when an external torque acts on it. When the torque is removed, the friction of the bearings and the air stops the wheel in 2.2 h.

'n Uniforme skyf roteer teen 'n konstante hoeksnelheid van 150 rev/min wanneer 'n eksterne draaimoment toegepas word daarop. Wanneer die draaimoment verwyder word, sal die wrywing van die laers en die lug veroorsaak dat die wiel in 2.2 uur tot stilstand kom.

- a) Calculate the constant angular acceleration of the disk during the slowdown.

Bereken die konstante hoekversnelling van die skyf terwyl dit tot stilstand kom.

[3]

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- b) How many revolutions does the disk make before stopping?
Hoeveel omwentelings maak die skyf voordat dit stop?

[3]

-
-
-
-
- c) At the instant the disk is turning at 75 rev/min, calculate the **tangential component** of the **linear** acceleration of a particle that is 50 cm from the axis of rotation.
*Die oomblik wat die skyf teen 75 rev/min draai, bereken die **raaklynige-komponent** van die **liniêre** versnelling van 'n deeltjie wat 50 cm van die rotasie as geleë is.*

[3]

ENJOY THE REST OF THE WEEKEND
GENIET DIE RES VAN DIE NAWEEK